Tzu-Ming Harry Hsu

徐子旻

Research scientist in embodied AI, computer vision, and learning systems h@stmharry.io

Computer vision, federated learning, sensing, and ML systems *+1 617-803-7785 / +886 928-494-198*

- MIT-trained researcher with 20+ publications, patents, and theses spanning computer vision, federated learning, sensing, and deployable medical AI.
- Built large-scale experimentation infrastructure at Google spanning 1,000+ GPUs and 1M+ machine hours, with open-source code and dataset releases.
- Research includes 3D scene understanding, wireless sensing, and learning under real-world data constraints, with outputs cited across academia and industry.
- Targeting embodied-AI roles that connect world models, robot perception, data systems, and real-world deployment.

Education

Massachusetts Institute of Technology Jun 2020 – Jun 2022
Ph.D., Computer Science *Cambridge, MA*

- *Advisor*: Peter Szolovits
- *Research areas*: Computer vision, federated learning, machine learning for healthcare

Massachusetts Institute of Technology Sep 2017 – May 2020
S.M., Electrical Engineering and Computer Science *Cambridge, MA*

- *Advisors*: Peter Szolovits, Fadel Adib
- *Research areas*: Computer vision, machine learning, wireless signal, signal processing

National Taiwan University Sep 2011 – Jun 2016
B.S., Physics; B.S.E., Electrical Engineering *Taipei, Taiwan*

- *Class rank*: 1/190
- *GPA*: 3.99 / 4.00

Professional Experience

Clarq AI Dec 2024 – Present
Co-founder & CTO *Palo Alto, CA / Taipei, Taiwan*

Scoped the data and modeling strategy for a sensor-rich human-performance platform, framing a long-term moat around force-time data, evaluation loops, and model-driven product decisions.

- Co-led technical positioning around a three-layer strategy: hardware wedge, SaaS profit engine, and long-term data/model moat.
- Drove US-facing investor outreach and progressed pre-seed conversations toward near-commit stage with angels.
- Supported Startup Island Taiwan Silicon Valley program participation and international expansion narrative.
- Aligned product roadmap, data strategy, and investor-facing story across Taiwan hardware operations and US capital market expectations.

International Academia of Biomedical Innovation Technology 2021 – Present
Consultant & Convener *Taipei, Taiwan*

Supported a 50+ member biomedical innovation community through project mentorship, dental AI research collaboration, and large-scale technology outreach events.

- Helped grow a community of 50+ members while supporting 8+ academic projects and 15+ publications.

- Led a dental AI research team in collaboration with National Taiwan University Hospital.
- Co-organized demo events with 100+ participants to showcase biomedical technologies and educate the broader community.

Dentscape Apr 2025 – Apr 2026
Machine Learning & Software Engineering Consultant *San Francisco, CA / Taipei, Taiwan*

Built data, training, and evaluation systems for AI-assisted dental design, combining geometric ML, pre-labeling workflows, and preference modeling to accelerate iteration and production deployment.

- Reorganized a fragmented ~10 TB dental dataset into a standardized ~4 TB ML-ready registry, reducing raw-data access from ~3 days to minutes.
- Rebuilt pre-labeling and labeling loops for 100-200 case batches, reducing turnaround from ~4 weeks to ~1 week and improving experiment velocity.
- Implemented bilevel/meta-learning for personalized crown generation and improved preference fit by ~20% while keeping the system production-facing.
- Optimized segmentation and crown-generation services for ~110% aggregate throughput improvement while reducing infrastructure cost by ~50%.

CodeGreen Labs PBC Sep 2023 – Nov 2024
Co-founder & CTO *Boston, MA / Taipei, Taiwan*

Led a 10-person engineering organization delivering auditable data systems for enterprise partners, reaching projected $1MM annual revenue while hardening integration, compliance, and delivery operations.

- Reached projected annual revenue of $1MM USD by Q2 2024 with a 10-person team.
- Established partnerships with Microsoft, Global Blockchain Business Council, Gold Standard, SDG Data Alliance, and PVBLIC.
- Built a cross-functional team across product, partnerships, and engineering to address complex data, auditability, and compliance challenges.
- Designed and shipped data-intensive systems with end-to-end auditability, regulatory compliance, and interoperability with existing partner ecosystems.

Hashgreen Labs Sep 2022 – Aug 2023
Co-founder & CTO *Boston, MA / Taipei, Taiwan*

Built and led a 20-person engineering organization, raised $1MM+ in capital, and shipped client-facing data and transaction systems while establishing hiring, delivery, and product operating cadence from zero.

- Raised over $1MM USD in capital and grew the team from 0 to 20 employees.
- Built hiring, onboarding, and performance review systems to support a high-performing engineering organization.
- Secured and delivered high-impact projects for clients including The World Bank and Chia Network.
- Led development of DeFi protocols (DEX and AMM), becoming the first team on the Chia blockchain to launch these primitives.

Massachusetts Institute of Technology Sep 2018 – Jun 2022
Research Assistant *Cambridge, MA*

Led research on deployable medical AI, computer vision, and learning under data constraints, producing 20+ publications, patents, and theses with translational partners across industry and healthcare.

- Published 14 conference articles, 5 journal articles, one U.S. patent, and two degree theses.
- Served as a peer reviewer for 8 conferences and journals, reviewing 34 articles.
- Received media coverage from CBS, Engadget, TechCrunch, and MIT News.
- Collaborated with Google, Takeda Pharmaceutical, and Beth Israel Deaconess Medical Center.
- Awarded scholarships totaling 4 years in duration.

WorldQuant Jun 2021 – Aug 2021
Data Science Intern *Taipei, Taiwan*

Improved an existing quantitative strategy by +1.5% market correlation through targeted modeling and optimization work.

Beth Israel Deaconess Medical Center Nov 2019 – Oct 2020
Research Trainee *Boston, MA*

Published impactful clinical AI research and deployed a PACS-integrated screening workflow across 1,000+ patients while managing shared deep-learning compute for a 10+ member research unit.

- Published three impactful research articles on critical medical topics, including liver tumor assessment, COVID-19 clinical risk prediction, and automatic pancreatic cancer assessment.
- Developed and implemented an AI-driven patient screening pipeline in the hospital's PACS workflow, screening over 1,000 patients and improving disease diagnosis accuracy.
- Managed the computation resources for a research unit of 10+ members, streamlining deep learning processes and maximizing resource sharing.

Google Jun 2020 – Sep 2020
Software Engineer Intern *Remote*

Contributed production software and experimentation tooling that improved reliability and iteration speed for applied machine learning workflows.

Google Jun 2019 – Mar 2020
Student Researcher *Seattle, WA / Cambridge, MA*

Built large-scale ML experimentation infrastructure spanning 1,000+ GPUs and 1M+ machine hours, and published 2 peer-reviewed papers in deep federated learning with open-source outputs for broader adoption.

- Conducted foundational research in deep federated learning, contributing to early growth of the field.
- Published 2 peer-reviewed papers and presented findings at top conferences and workshops.
- Developed a scalable experimentation library running on thousands of machines and 1,000+ GPUs (over 1M machine hours).
- Open-sourced research code and datasets for broader community use.

Digital Drift Mar 2016 – Mar 2020
Lead Machine Learning Scientist *Taipei, Taiwan*

Built and deployed deep learning systems and backend APIs that moved computer vision features from experimentation into production mobile experiences.

- Built deep neural networks for cuisine recognition and matching using TensorFlow on multi-GPU machines.
- Designed and maintained backend APIs to serve model predictions in production settings.

Brigham and Women's Hospital Mar 2019 – Feb 2020
Research Trainee *Boston, MA*

Contributed hospital-facing medical-imaging ML work on risk prediction and deployment-oriented evaluation using real patient cohorts and radiology workflows.

- Supported model development and validation analysis for clinical imaging applications tied to operational hospital use cases.

Academia Sinica Feb 2014 – May 2016
Student Researcher *Taipei City, Taiwan*

Researched heterogeneous, unsupervised, and semi-supervised domain adaptation for visual classification and published three peer-reviewed papers presented at top conferences and workshops.

- Researched heterogeneous, unsupervised, and semi-supervised domain adaptation for visual classification.
- Published 3 peer-reviewed academic papers and presented findings at top conferences and workshops.

Olympiad Tutoring Community Sep 2011 – Jun 2015
Private Tutor *Taipei, Taiwan*

Mentored advanced physics students from fundamentals through competition-level performance, including pathways to Taiwan's national Olympiad representation.

- Mentored students who became Taiwan national representatives in the International Physics Olympiad (IPhO).

Leadership Experiences

Federation of Taiwanese Student Associations in New England (FT-SANE) 2019 – 2020
Activities Officer *Boston, MA*

MIT Taiwanese Student Association 2018 – 2019
President *Boston, MA*

- Led coordination of 20+ events for a diverse community of over 100 members.
- Organized 10+ career workshops and helped plan 10+ national holiday celebrations.

Awards & Honors

Departmental Valedictorian (First Place at Graduation) 2016
Department of Electrical Engineering, National Taiwan University *Taipei, Taiwan*

- Placed 1st among 200 classmates across all courses.

Silver Medal 2015
Altera Innovate Asia FPGA Design Competition *Taipei, Taiwan*

- Placed 2nd among 20 international teams.
- Designed a product named EZBud with integrated algorithms that modulate music based on user sporting statistics.

Gold Medal / Overall Winner 2011
42nd International Physics Olympiad (IPhO) *Bangkok, Thailand*

- Represented Taiwan and placed 1st in theory, experiment, and combined rankings among 400+ participants from 80+ countries.

Gold Medal / Experiment Winner 2011
12th Asian Physics Olympiad (APhO) *Tel Aviv, Israel*

- Placed 1st in the experiment section among 100+ participants.

Gold Medal 2010
41st International Physics Olympiad (IPhO) *Zagreb, Croatia*

Gold Medal / Experiment Winner 2010
11th Asian Physics Olympiad (APhO) *Taipei, Taiwan*

- Placed 1st in the experiment section among 100+ participants.

Gold Medal 2008
5th International Junior Science Olympiad (IJSO) *Changwon, Korea*

Artificial Intelligence to Assess Dental Findings from Panoramic Radiographs—A Multinational Study 2025
arXiv *Cited by 8*

Yin-Chih Chelsea Wang, Tsao-Lun Chen, Shankeeth Vinayahalingam, Tai-Hsien Wu, Chu Wei Chang, Hsuan Hao Chang, Hung-Jen Wei, Mu-Hsiung Chen, Ching-Chang Ko, David Anssari Moin, Bram van Ginneken, Tong Xi, Hsiao-Cheng Tsai, Min-Huey Chen, **Tzu-Ming Harry Hsu**, Hye Chou

Intra-Oral Scan Segmentation Using Deep Learning 2023
BMC Oral Health *Cited by 42*

Shankeeth Vinayahalingam, Steven Kempers, Julian Schoep, **Tzu-Ming Harry Hsu**, David Anssari Moin, Bram van Ginneken, Tabea Flugge, Marcel Hanisch, Tong Xi

DeepOPG: Improving Orthopantomogram Finding Summarization with Weak Supervision 2021
Medical Image Computing and Computer Assisted Intervention (MICCAI 2021) *Cited by 8*

Tzu-Ming Harry Hsu, Yin-Chih Chelsea Wang

Federated Visual Classification with Real-World Data Distribution 2020
European Conference on Computer Vision (ECCV 2020) *Cited by 275*

Tzu-Ming Harry Hsu, Hang Qi, Matthew Brown

Measuring the Effects of Non-Identical Data Distribution for Federated Visual Classification 2019
arXiv Preprint arXiv:1909.06335 *Cited by 1972*

Tzu-Ming Harry Hsu, Hang Qi, Matthew Brown

3D-Aware Scene Manipulation via Inverse Graphics 2018
Advances in Neural Information Processing Systems (NeurIPS 2018) *Cited by 321*

Shunyu Yao, **Tzu Ming Hsu**, Jun-Yan Zhu, Jiajun Wu, Antonio Torralba, Bill Freeman, Josh Tenenbaum

Learning Food Quality and Safety from Wireless Stickers 2018
Proceedings of the 17th ACM Workshop on Hot Topics in Networks (HotNets 2018) *Cited by 95*

Unsoo Ha, Yunfei Ma, Zexuan Zhong, **Tzu-Ming Hsu**, Fadel Adib

Transfer Neural Trees: Semi-Supervised Heterogeneous Domain Adaptation and Beyond 2019
IEEE Transactions on Image Processing *Cited by 21*

Wei-Yu Chen, **Tzu-Ming Harry Hsu**, Yao-Hung Hubert Tsai, Ming-Syan Chen, Yu-Chiang Frank Wang